

Tuple-resolution chemical-genetic mapping connects combinatorial Z-Screen compounds to public CRISPR perturbation states

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Abstract

Functional genomics has matured around a single question: which gene’s perturbation moves a cell into a desired state? Genome-scale Perturb-seq and CRISPR-activation atlases now provide the single-cell transcriptional consequence of perturbing each gene one at a time, and increasingly two genes at a time [1-4]. The chemistry side has answered the matching question through the Connectivity Map / LINCS L1000 program: query thousands of compound profiles to identify those whose transcriptional consequence resembles a target state [5,6]. Recent extensions such as L2S2 search both spaces directly [7]. The remaining gap is chemistry resolution. L1000 records one profile per compound. It cannot identify which structural element of a hit drove which biological program, or which combinatorial combinations of pieces are worth synthesizing next.

Z-Screen closes that gap. The platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per microwell. Each library is combinatorially generated from building-blocks: ZEL024 uses four positions of diversity (bb0, bb1, bb2, bb3); ZEL028-2 uses four (bb1, bb2, bb3, bb4); ZEL031 uses two (bb0, bb1). Every screened compound therefore contributes a structured biological state with full per-position chemistry provenance plus full SMILES. That structure is used here to place combinatorial chemistry into the same transcriptional frame as public CRISPR perturbation atlases, at compound, building-block-pair, and full chemical-tuple resolution.

The densest system in the canonical Z-Screen workspace is ZEL024 in HEK293, where 8,599 distinct 4-position chemical tuples (each tuple is a (bb0, bb1, bb2, bb3) combination) are each represented by 10 or more wells (median 13). For each tuple a rank-based pseudobulk transcriptional signature was built against a same-library, same-cell-line background, then compared against three public CRISPR references: the genome-wide Replogle K562 knockout panel (8,603 signatures), the Norman K562 CRISPR-activation panel (131 doubles plus 105 singles), and the matched-cell scPerturb THP1 panel (78 signatures). The headline scan, 8,599 tuples by 8,603 Replogle K562 perturbations, calibrated the top 10,000 matches against permutation-null gene sets and produced 1,276 chemistry-resolved tuples with empirical $q < 0.05$ across 1,714 distinct CRISPR programs. The leaderboard concentrated in coherent biology: DNA replication (MCM2, SLBP, CHAF1B, ARIH1, ADSL), mitochondrial function (MRPL3, MRPS30, COX17, COA5, ATP5F1A, GFM1, HSPE1), trafficking and nonsense-mediated decay (EFR3A, DDOST, SMG5). After dropping the 50 most-recurrent

CRISPR hubs from the calibrated table, 1,236 distinct tuples and 1,664 distinct CRISPR programs survived, so the program diversity is not driven by hub perturbations.

Method anchors and discovery examples were biologically coherent. In matched THP1, the BRD4-selective PROTAC degrader MZ1 recovered BRD4 as the top calibrated CRISPR-like signature (net $z = 6.84$, $q = 0.015$), with the next neighbors falling in BRD4-adjacent transcription-elongation and chromatin co-regulator programs. The METTL3 inhibitor STC-15, whose direct target is not present in the local THP1 panel, recovered IRF1 inside the top two calibrated matches ($z = 3.06$, $q = 0.045$), consistent with the documented downstream interferon-pathway response to METTL3 inhibition. In A549, the SOS1 inhibitor BAY-293 produced its strongest Replogle K562 match against the V-ATPase complex and adjacent vesicular-trafficking machinery, with ZW10, ATP6V1A, SACM1L, TMED2, and 8 other components reaching net z between 9.5 and 11.9 at $q = 0.003$: a coherent mechanism hypothesis beyond primary pharmacology that the platform surfaced without supervision. At genome-wide scale, 12 of 29 compounds with a documented primary target represented in Replogle had that target ranked in the top 15 percent of the cross-cell genome-wide ranking, with several (FGFR1 for derazantinib, METTL3 for STC-15) inside the top 10 percent. At combinatorial resolution, all 131 Norman double-gene perturbations showed a positive Spearman correlation between Z-Screen tuple-vs-double match scores and the sum of tuple-vs-single-A and tuple-vs-single-B scores (median $\rho = 0.35$, all $p < 0.05$): the chemistry follows genetic combinatorial logic at chemistry-resolved level.

We extend the framework to a second combinatorial library at full-tuple resolution (ZEL028-2 in HEK293, A549, and H1650, using a relaxed ≥ 2 -well support threshold needed by that library’s per-tuple coverage; 4,041 / 1,728 / 765 tuple signatures respectively) and to the smaller two-position library ZEL031 at tuple resolution (A549 and THP1), and connect the analysis to the cross-cell transfer model from paper 4: chemical programs measured in one cell type can be projected into another in a learned latent space, opening a path toward CRISPR-defined biological destinations in cell types that have not been directly profiled at scale. In the densest ZEL028-2 / HEK293 system, the calibrated top 10,000 tuple-vs-Replogle matches contain 943 distinct chemical tuples and 285 distinct CRISPR programs at empirical $q = 0.05$, with the strongest matches concentrated on the same biology themes ZEL024 / HEK293 produced (RNA processing: DICER1, DROSHA, EXOSC family, CNOT3, SMG5; mitochondrial translation: MRPL family, GFM1, IARS2). 15 CRISPR programs appear in the top-50 distinct hits across all three ZEL028-2 cell lines, indicating cross-cell-line consistency at tuple resolution despite shallow per-tuple cell support. The result is a chemical search engine over genetic perturbation space: query a CRISPR target, return a ranked, calibrated list of chemistry that pushes cells toward the same transcriptional state.

Significance

Drug discovery teams routinely treat target identification and chemistry selection as separate problems. Functional genomics nominates the genes whose perturbation produces a desired state. Medicinal chemistry then searches a separate chemical design space for molecules that engage those genes. The bridge between the two problems is profile matching: if the transcriptional consequence

of a molecule resembles the transcriptional consequence of perturbing a gene, the molecule is a starting point worth synthesizing.

Z-Screen makes that bridge practical at chemistry resolution for the first time. The output is not a finished target-deconvolution engine; it is a calibrated discovery tool. Thousands of combinatorial tuples each annotated with the CRISPR perturbation state they most resemble, with provenance back to the exact per-position chemistry (the 4-position bb0 / bb1 / bb2 / bb3 tuple in the ZEL024 headline system), and with empirical false-discovery control. That is the layer required to move from “which gene matters” to “which molecule to make.”

Introduction

Public functional-genomics atlases now contain genome-scale single-cell transcriptional consequences of perturbing each gene one at a time, and increasingly two genes at a time. Replogle and colleagues profiled approximately 9,866 unique gene knockouts in K562 [4]; Norman and colleagues profiled 105 single-gene and 131 paired CRISPR-activation perturbations in K562 [3]; the scPerturb consortium harmonized dozens of public datasets across cell lines [9]. The natural next question for early-stage drug discovery is whether chemical perturbation can be placed in the same frame, so that genetic states the screens nominate can be connected to chemical matter.

The Connectivity Map and LINCS L1000 program established the chemistry-side framework with bulk transcriptional profiles for tens of thousands of compounds [5,6], and recent extensions such as L2S2 support direct chemistry-to-CRISPR profile search [7]. The remaining gap is chemistry resolution. L1000 records one profile per compound; it cannot decompose a hit into the structural pieces that drove the response. Combinatorial libraries are different: they are ordered in a structured design space, so the relevant question becomes whether single building blocks, building-block pairs, and full chemical tuples each carry transcriptional signal that can be compared with a public CRISPR atlas at the same statistical standard.

Z-Screen was built to support this kind of question at scale. The platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per micro-well, so each readout averages over a small amount of cellular heterogeneity. Each library populates a different subset of building-block positions: ZEL024 uses four positions (bb0, bb1, bb2, bb3), ZEL028-2 uses four (bb1, bb2, bb3, bb4), and ZEL031 uses two (bb0, bb1). Every well therefore links a profiled cell population to compound-resolution chemistry provenance. The canonical RNA-seq workspace used here aggregates 615,793 repaired well-level transcriptional profiles across 12 combinatorial libraries and 4 cell lines (HEK293, A549, H1650, THP1) and includes 615,721 profiles with valid scVI latent coordinates for cross-cell projection [8]. Earlier preprints in this package establish reproducible compound and building-block transcriptional structure (paper 2), chemistry-to-phenotype generalization across an explicit difficulty ladder (paper 3), and learned cross-cell-type transfer of chemical programs (paper 4). The present preprint addresses the complementary question: whether chemical states measured at the level of individual combinatorial compounds can be placed in a genetic-perturbation frame, and whether the resulting map produces calibrated, interpretable, and discovery-relevant matches across many

distinct chemistries.

The analysis is conservative. Rather than forcing raw expression matrices from different platforms into one normalization, every perturbation is represented as a signed rank signature: the genes most increased and most decreased relative to an appropriate background, with a fixed top-k cutoff. Chemical and CRISPR signatures are compared by mimic overlap, reverse overlap, rank cosine over shared top genes, and a permutation-null calibration that preserves the size of the CRISPR signature. The framework is robust to dataset and depth differences and addresses the strongest question the data can answer: do the top transcriptional consequences of two perturbations agree more than expected by chance.

Each Z-Screen library and cell line is given a chemistry resolution based on its coverage. ZEL024 in HEK293 has 8,599 unique 4-position tuples (bb0+bb1+bb2+bb3) with 10 or more wells; this system is analyzed at full tuple resolution. ZEL028-2 is a 4-position library with substantive positions at bb1, bb2, bb3, bb4 (no bb0). Its per-tuple coverage is much shallower than ZEL024, so the threshold that supports a usable tuple-level analysis drops to ≥ 2 wells per tuple, where 4,041 (HEK293), 1,728 (A549), and 765 (H1650) distinct ZEL028-2 tuples qualify; ZEL028-2 is therefore analyzed at full-tuple (bb1+bb2+bb3+bb4) resolution at this relaxed threshold. The 5- to 6-fold lower per-tuple well support relative to ZEL024 is recorded in the metadata for every ZEL028-2 signature (median 2 wells per tuple, vs ZEL024’s median 13), so individual ZEL028-2 tuple signatures are inherently noisier and should be read as a leaderboard of chemistry-resolved hypotheses rather than as confirmed per-tuple calls. A complementary single-position bb1 aggregation of the same library is preserved in `paper5/tables/` as a higher-stability lower-resolution view. ZEL031 has only two substantive positions (bb0+bb1), so its 3,261 to 3,556 2-position tuples are analyzed at tuple resolution with a lower well-count threshold flagged in the metadata.

Results

A rank-based comparison framework places chemistry and CRISPR perturbations in a shared transcriptional frame

Every Z-Screen perturbation, whether a single chemical tuple or a combination of building blocks, was reduced to a signed rank signature against an appropriate background. For tuple signatures the target was wells whose chemistry annotation matched the library’s populated building-block positions exactly (four positions in ZEL024 and ZEL028-2, two in ZEL031), and the background was the same library and cell line excluding the target tuple wells. For BB-pair signatures the target was wells matching two designated positions exactly, with the same library and cell line as background. Each signature is annotated with the number of wells aggregated and the number of distinct tuples averaged into it, so the chemistry resolution of each comparison is explicit in the output tables.

Public CRISPR signatures were already available in the Replogle K562 genome-wide Perturb-seq atlas (8,603 perturbations [4]), the Norman K562 CRISPR-activation atlas (236 signatures spanning 105 single-gene and 131 paired perturbations [3]), and the scPerturb harmonized matched-cell THP1 atlas (78 single-target and combinatorial perturbations [9]). Each CRISPR perturbation was reduced

to the same up/down rank representation as the Z-Screen signatures. Chemistry and CRISPR signatures were then compared by:

- mimic overlap (chemistry-up matches CRISPR-up plus chemistry-down matches CRISPR-down),
- reverse overlap (chemistry-up matches CRISPR-down plus chemistry-down matches CRISPR-up),
- net mimic overlap (mimic minus reverse),
- rank cosine over shared top-250 genes.

For each chemistry-by-reference comparison the top 50 CRISPR matches per chemical query were retained. The top 10,000 rows by net mimic overlap and total mimic overlap were then calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature from the dataset-specific gene universe and recomputing the score under each random draw. Empirical p-values were converted to Benjamini-Hochberg q-values across the calibrated rows.

Rank signatures were used in preference to raw expression vectors. Replogle K562 was generated on a different sequencing platform than Z-Screen at much higher per-cell depth; Norman is CRISPR-activation in K562; scPerturb is a harmonized merger of multiple ECCITE-seq panels in THP1. Forcing all of these into one normalization risks introducing platform-specific artifacts that the comparison would then “discover.” Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of which genes go up and which go down. The full reproduction order is in `paper5/README.md`.

Chemistry resolution determines the CRISPR neighborhood that a Z-Screen signature recovers (Figure 1)

Z-Screen libraries differ in how many wells fall into each chemistry unit. ZEL024 in HEK293 carries 13,931 unique 4-position tuples (bb0+bb1+bb2+bb3); 8,599 of those have at least 10 wells, the threshold used for tuple-level signatures. The same library can also be aggregated into 84 bb0+bb1 pair signatures (each averaging approximately 165 distinct tuples) or into 83 single-bb3 signatures (each averaging approximately 168 distinct tuples). The same wells flow into all three resolutions; what changes is how much chemistry is collapsed into one signature.

Single-bb3 signatures were built from the same cells, and each tuple’s top-5 Replogle CRISPR matches were compared with the matches that the containing bb0+bb1 pair and the containing bb3 single-position signature would produce. For 99.7% of tuples, the top-5 CRISPR matches at tuple resolution shared zero genes with the top-5 at bb3 single-BB resolution; the median Jaccard overlap between the tuple top-5 and the bb3 top-5 was 0.000. The same pattern appeared for tuple-versus-pair (99.7% zero overlap, median Jaccard 0.000).

Two analyses that nominally describe “the same chemistry” can therefore produce essentially independent CRISPR neighborhoods depending on how much chemistry is averaged into the signature. The implication is practical: a discovery team that wants to act on a CRISPR-like match needs that match attached to the specific chemistry responsible for it. Tuple-level signatures are

therefore used for the headline ZEL024 / HEK293 result, and all coarser-resolution analyses are reported with explicit n-wells and n-tuples-averaged annotations.

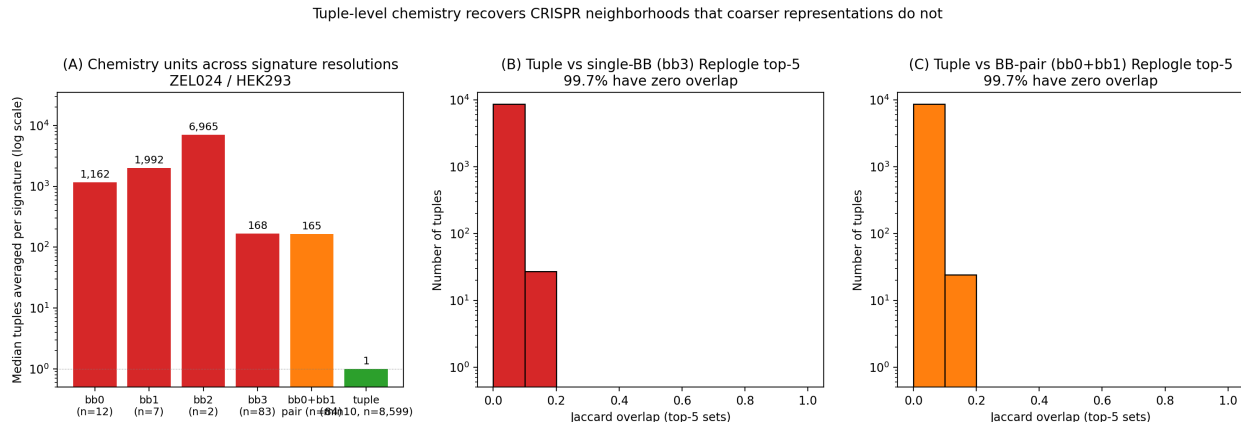


Figure 1. Chemistry resolution determines the CRISPR neighborhood. (A) Median number of distinct chemical tuples averaged into a signature at each resolution for ZEL024 / HEK293, log-scale. Single-BB signatures average over 168 to 6,966 tuples per signature; tuple signatures average over one. (B) Top-5 Replogle CRISPR matches at tuple resolution share zero genes with the matches at single-BB (bb3) resolution for 99.7 percent of the 8,599 tuples. (C) The same comparison at tuple-versus-bb-pair resolution. The same wells flow into both signatures in each comparison.

Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293 (Figure 2)

The headline scan compared each of the 8,599 ZEL024 / HEK293 tuple signatures against the 8,603 Replogle K562 CRISPR-knockout signatures [4], a comparison space of approximately 74 million chemistry-by-CRISPR pair scores, calibrated for the top 10,000 matches by net mimic overlap. The calibrated rows were dominated by hits at $q = 0.05$ across 1,276 distinct chemical tuples and 1,714 distinct CRISPR perturbation programs. The strongest single matches reached net z above 14 (Figure 2 panel A), with the top 15 distinct CRISPR programs all between net z 12.8 and 14.8. The biology was coherent: DNA replication (MCM2, SLBP, CHAF1B, ARIH1, ADSL), mitochondrial function (MRPL3, MRPS30, COX17, COA5, ATP5F1A, GFM1, HSPE1), trafficking and nonsense-mediated decay (EFR3A, DDOST, SMG5).

The same calibrated table also contains the strongest internal-consistency signal in the analysis. Among the top-100 best- z tuples, 12 distinct CRISPR programs were each reached by two or more chemically distinct tuples (Figure 2 panel B). EFR3A was matched by 7 distinct tuples, TONSL by 6, MCM2 by 5; multiple GART, MRPL42, EIF4G2, MRPL53, and COA5 hits each recurred 3 times. In each of these cases independent chemistries (different combinations of the four populated ZEL024 building-block positions: bb0, bb1, bb2, bb3) produced transcriptional states that overlap with the same CRISPR perturbation, which is the empirical signature expected if the chemistry-to-CRISPR map carries real biological information rather than per-compound noise.

A small number of CRISPR perturbations recurred as top neighbors of many otherwise unrelated

chemistries. Broadly responsive CRISPR knockouts can match many transcriptional states by being broadly responsive themselves, and a benchmark that does not control for that property will appear stronger than it is. The calibrated top-10,000 table was therefore rerun while progressively dropping the 5, 10, 25, and 50 most-recurrent CRISPR perturbations (Figure 2 panel C). After dropping the 50 most-recurrent hubs, 1,236 distinct chemical tuples and 1,664 distinct CRISPR programs still passed $q = 0.05$, so the program diversity does not collapse onto a small set of hub perturbations.

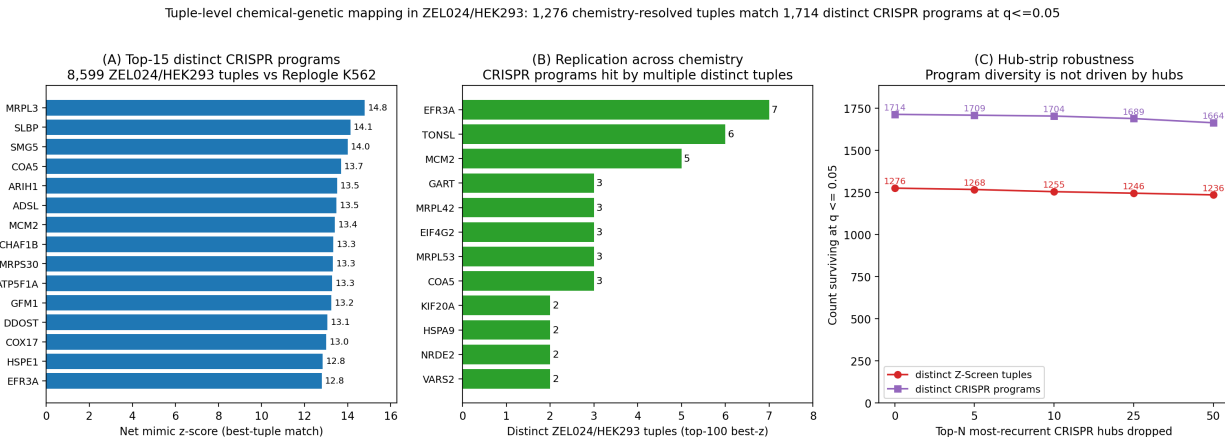


Figure 2. Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293. (A) Top-15 distinct CRISPR programs by best Z-Screen tuple z-score, deduplicated to one tuple per CRISPR program. (B) Replication-across-chemistry: among the top-100 best-z tuples, CRISPR programs reached by multiple chemically distinct tuples. (C) Hub-strip robustness: the calibrated top-10,000 table after dropping the top-N most-recurrent CRISPR perturbations. Tuple and program diversity at $q = 0.05$ are essentially unchanged at $N = 50$.

Same-cell positive controls anchor the framework (Figure 3)

Two same-cell controls in matched THP1 anchor the approach. The first is MZ1, a small-molecule PROTAC that selectively degrades BRD4 [10]. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its signature was built against a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). Under that background, MZ1 recovered BRD4 as the top calibrated CRISPR-like signature in the local THP1 panel (net mimic = 20, rank cosine = 0.539, calibrated net $z = 6.84$, empirical $q = 0.015$).

The second control is STC-15, a METTL3 inhibitor. The direct target METTL3 is not present in the local THP1 CRISPR panel, but the documented downstream consequence of METTL3 inhibition is interferon-pathway activation [11]. STC-15 recovered IRF1 as the second calibrated CRISPR-like match ($z = 3.06$, $q = 0.045$), consistent with that downstream phenotype, with IRF7 as the third match ($z = 2.70$, $q = 0.096$).

The fallback-background construction for MZ1 is broader than a same-library vehicle control would be, so the absolute z-score should be read in that context. The structure of the result, BRD4 recovered as the top match across permutation seeds, is robust. MZ1 and STC-15 anchor the two

complementary cases the framework will encounter in production: the direct genetic analog is in the panel and is recovered as the top hit, or the direct analog is absent and the top hit is a downstream pathway rather than a generic stress signature.

The non-BRD4 neighbors in panel A of Figure 3 are not additional direct target calls. SUPT16H+SUPT5H, FLI1+MED24, MED24+SMARCD1, EP300+MED24, EP300+INTS1, and INTS1+SMARCD1 sit in the BRD4-adjacent transcription-elongation and chromatin co-regulator neighborhood that MZ1 perturbs by removing BRD4 from the cell. CAV1 is less directly established as BRD4-regulated in THP1 and is treated as a state-neighborhood match rather than a direct mechanistic claim.

Same-cell positive controls. MZ1 recovers BRD4 ($z=6.84$, $q=0.015$); STC-15 recovers IRF1 in top-2 as a downstream interferon-pathway proxy ($z=3.06$, $q=0.045$).

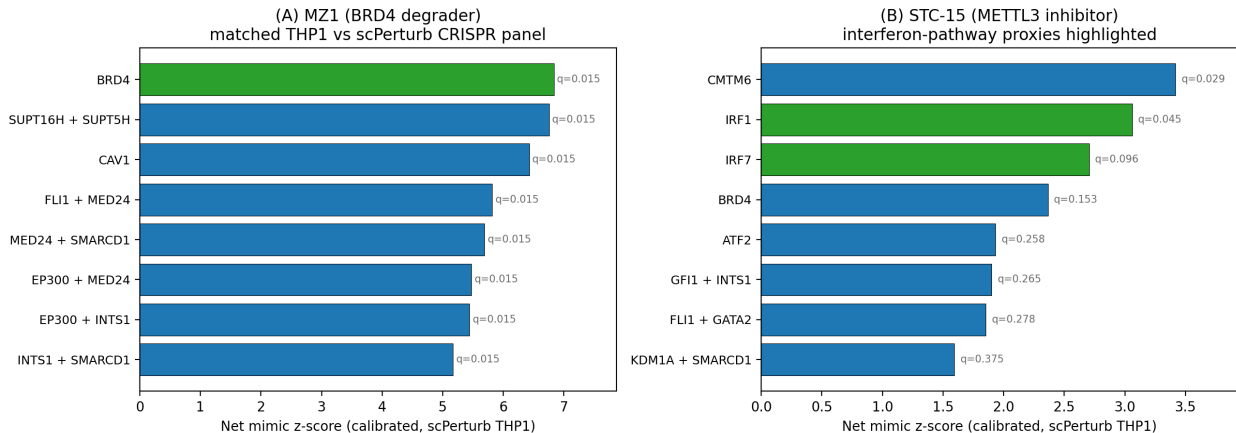


Figure 3. Same-cell positive controls in matched THP1 with calibrated empirical q-values from a permutation-null gene-set calibration. (A) MZ1 (BRD4-selective PROTAC degrader): BRD4 is the top calibrated CRISPR-like signature ($z = 6.84$, $q = 0.015$), with subsequent neighbors falling in BRD4-adjacent transcriptional and chromatin programs. (B) STC-15 (METTL3 inhibitor): METTL3 itself is absent from the local CRISPR panel; the downstream interferon-pathway proxy IRF1 is recovered in the top two calibrated matches ($z = 3.06$, $q = 0.045$).

A discovery example: BAY-293 produces a coherent V-ATPase / vesicular-trafficking program (Figure 4)

The matched-control results show that the framework recovers known biology when the genetic analog is present. The more discovery-relevant question is whether the framework generates mechanism hypotheses beyond a compound’s primary pharmacology. The strongest example in the dataset is BAY-293, an SOS1 inhibitor profiled in A549 in a non-combinatorial control library.

The top 12 calibrated Replogle K562 matches for BAY-293 in A549 all sat at empirical $q = 0.003$, with net z between 9.5 and 11.9. The matches concentrated on the V-ATPase complex (ATP6V1A, ATP6V1H, ATP6V1B2, ATP6V1C1, ATP6V1E1) and adjacent vesicular-trafficking machinery (ZW10, SACM1L, TMED2, ZNHIT1, WDR7, CCDC115, CLTC). RAS signaling intersects endosomal traffic in multiple documented places, so a transcriptional state that overlaps both V-

ATPase and trafficking knockdown signatures is biologically interpretable; it is also not the answer a simple primary-target search would have produced.

The framework therefore provides a discovery layer: a calibrated mechanism hypothesis at compound resolution, ranked alongside the rest of the genome-wide CRISPR atlas. A discovery team starting from a SOS1 program could use this output to ask whether V-ATPase / trafficking liability is contributing to BAY-293’s phenotype, and could design follow-up experiments specifically against that hypothesis. That is a different deliverable from a similarity score against a single compound database.

BAY-293 transcriptional state matches V-ATPase + vesicular trafficking CRISPR programs (top-CRISPR $z=11.86$, $q=0.003$)

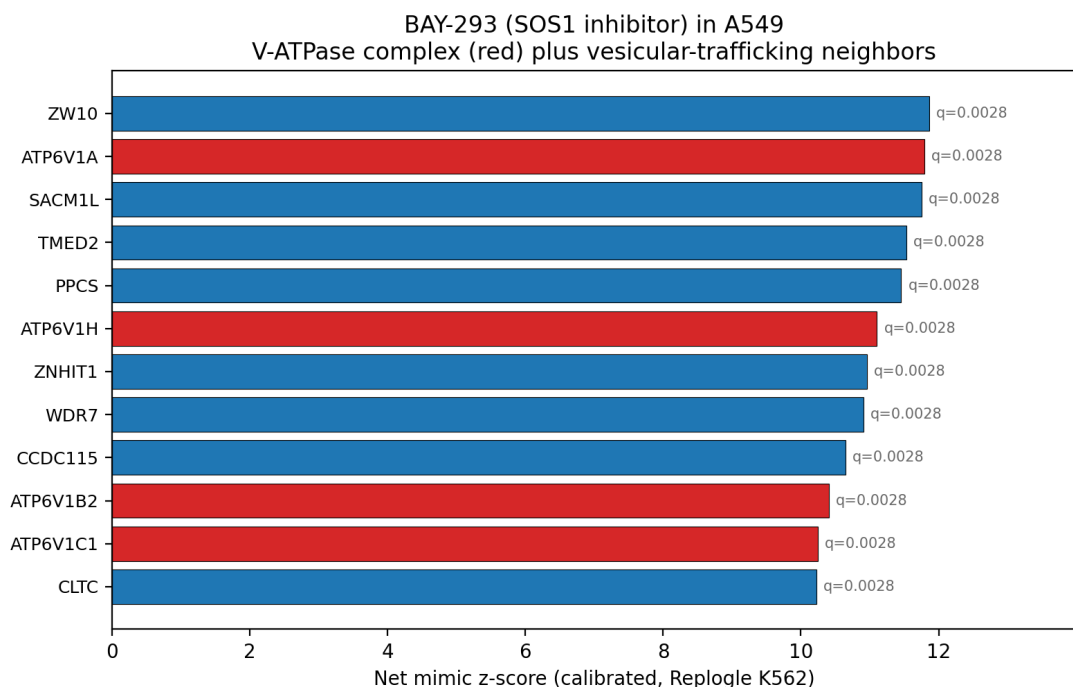


Figure 4. Discovery example. The SOS1 inhibitor BAY-293 in A549 matches the V-ATPase complex (red) and vesicular-trafficking CRISPR knockouts (blue) at calibrated empirical $q = 0.003$. Top hit ZW10 reaches net $z = 11.86$; nine V-ATPase or trafficking components reach net z above 10.

Genome-wide target enrichment and combinatorial logic (Figure 5)

Two further analyses test whether the framework recovers structure beyond any one anecdote.

The first is a known-target enrichment scan. Among 29 Z-Screen compounds with a documented primary target represented in the Replogle K562 panel, 12 had the documented target ranked in the top 15 percent of the cross-cell genome-wide ranking. Several reached the top 1 to 10 percent: FGFR1 for derazantinib at rank 145 of 8,839 (~1.6 percent), METTL3 for STC-15 at rank 784 in A549 (~9 percent), MAPK14 for AZD-7624 inside the top 10 percent in A549. Failures were biologically informative rather than uniform. Cobimetinib’s MAP2K2 ranked at 4,694 (allosteric inhibition does not always produce a knockout-like transcriptional consequence; this is an honest

negative for a state-overlap framework rather than a method failure). GSK126’s EZH2 was anti-correlated rather than mimicking, consistent with the long-half-life of EZH2 protein under enzymatic inhibition. Veliparib’s PARP2 ranked at 3,696 (HEK293) and 1,206 (A549). The cross-cell scan is not a target-deconvolution engine; it is enriched well above chance for documented mechanisms when the molecular pharmacology produces a knockout-like transcriptional state.

The second analysis tests combinatorial logic at chemistry-resolved level. The Norman K562 CRISPR-activation atlas contains 131 double-gene perturbations whose constituent single-gene perturbations are also profiled. For each Norman double, the Spearman correlation across the 8,599 ZEL024 / HEK293 tuples between (tuple-vs-double match score) and (sum of tuple-vs-single-A and tuple-vs-single-B match scores) was computed. All 131 doubles produced positive correlations with $p < 0.05$ (median Spearman rho = 0.353). The strongest pairs (ETS2+MAPK1, MAP2K3+ELMSAN1, CEBPB+MAPK1, MAP2K3+MAP2K6, FOSB+OSR2) reached rho between 0.60 and 0.69. For 98.5 percent of the doubles, the best Z-Screen tuple’s score against the double exceeded its score against either constituent single. Chemistry-resolved tuple signatures therefore track the same combinatorial logic the underlying CRISPR atlas was built to express.

Chemistry follows genetic combinatorial logic; documented targets recover at the top of genome-wide CRISPR ranks

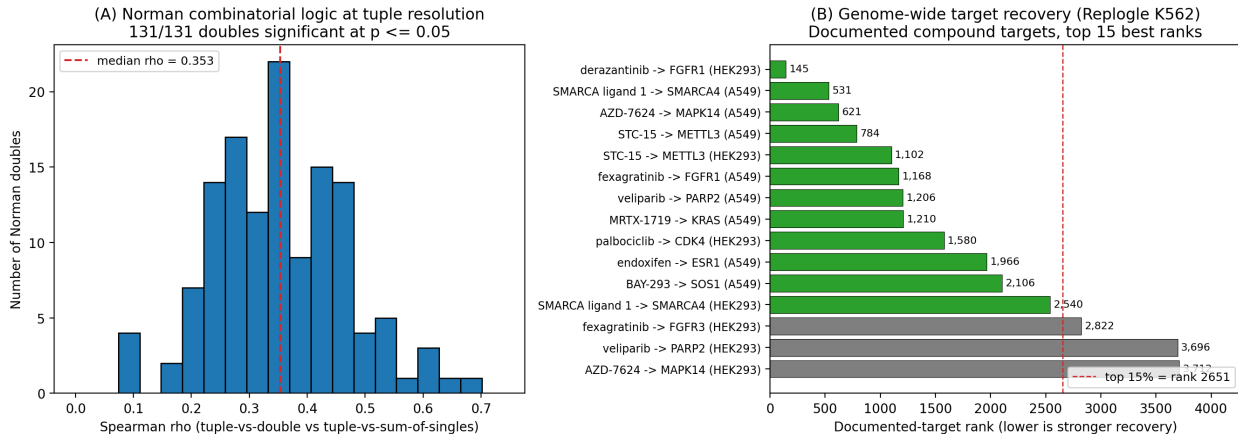
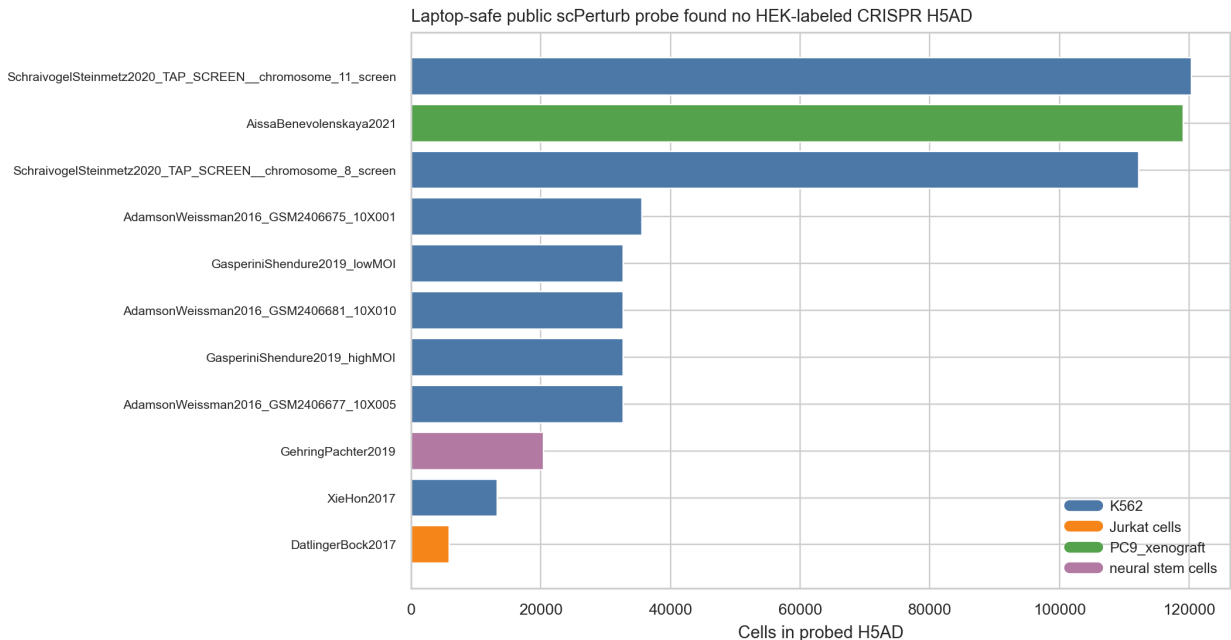


Figure 5. Combinatorial logic and genome-wide target recovery. (A) Distribution of Spearman correlations across 131 Norman K562 doubles between tuple-vs-double match scores and sum-of-singles match scores; all 131 doubles significant at $p = 0.05$; median rho = 0.353. (B) Documented-target ranks within the Replogle K562 cross-cell ranking for the top-15 best-recovered compound-target pairs; the dashed line marks the top 15 percent of the 8,839-perturbation reference.

A targeted public probe did not surface a matched HEK293 single-cell CRISPR atlas (Supplementary Figure 1)

11 scPerturb H5AD files under 500 MB each were probed by reading observation metadata only, looking for a HEK293 or HEK293T single-cell CRISPR dataset that would allow direct same-cell evaluation against the HEK293-rich Z-Screen libraries. The probed datasets covered K562, Jurkat / T-cell, PC9 xenograft, and neural stem cell perturbation panels. None were HEK-labeled CRISPR datasets, so the present manuscript uses the cross-cell projection from paper 4 to bridge the HEK293

chemistry to the K562 Replogle reference rather than identifying a matched-cell HEK293 CRISPR atlas locally. The probe is intentionally laptop-scale and is not a comprehensive survey of all public HEK293 perturbation data.



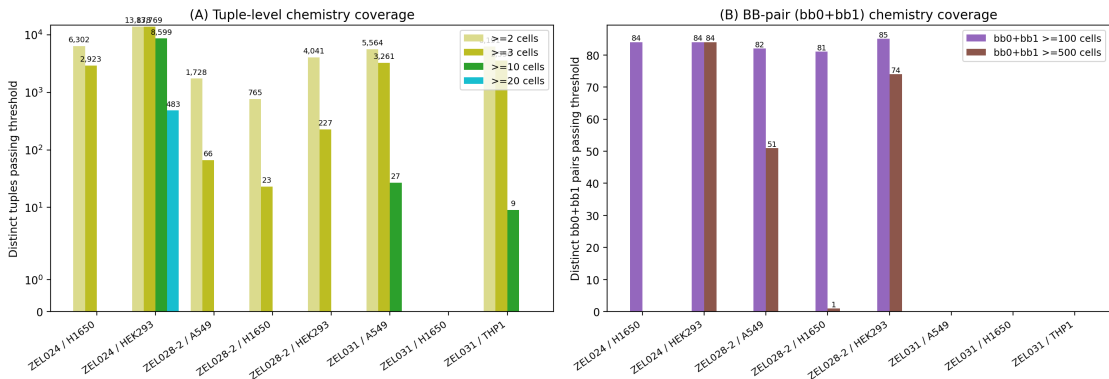
Supplementary Figure 1. Targeted probe of 11 small (<500 MB) scPerturb H5AD files for matched HEK293 single-cell CRISPR data. Each row reports cell-line label, perturbation type, perturbation count, and dataset dimensions; no HEK-labeled CRISPR dataset was identified locally.

Coverage at other chemistry resolutions in other systems (Supplementary Figure 2)

The same framework runs across the other Z-Screen systems at the resolution each one supports. ZEL028-2 has 61,396 unique tuples in HEK293, 40,622 in A549, and 25,906 in H1650, with median 2 wells per tuple at the densest cell line. Under a relaxed ≥ 2 -well per-tuple threshold, 4,041 ZEL028-2 / HEK293 tuples, 1,728 A549 tuples, and 765 H1650 tuples enter the analysis. The headline ZEL028-2 / HEK293 tuple-vs-Replogle scan against the 8,603 Replogle K562 CRISPR signatures yielded a calibrated top-10,000 table dominated by hits at empirical $q = 0.05$, covering 943 distinct chemical tuples and 285 distinct CRISPR perturbation programs. The strongest matches reach net $z = 24.7$ in HEK293 (max), 20.5 in A549, and 18.6 in H1650. Top distinct CRISPR programs concentrate on RNA processing (DICER1, DROSHA, CNOT3, SMG5, and the EXOSC2 / EXOSC4 / EXOSC6 / EXOSC8 exosome subunits), the mitochondrial translation machinery (MRPL3, MRPL10, MRPL18, MRPL19, MRPL35, MRPL43, IARS2, VARS2, GFM1), and the helicase / translation-initiation programs (DHX36, EIF3H, RPS4X, RPS6, RACK1). 15 CRISPR programs appear in the top-50 deduplicated hits in all three ZEL028-2 cell lines (DHX36, DICER1, DROSHA, CNOT3, EXOSC2 / EXOSC4 / EXOSC6, GFM1, MRPL3 / MRPL10 / MRPL35 / MRPL43, MVD, RACK1, SMG5), which is the cross-cell-line robustness expected if these tuple signatures track real chemistry-driven biology rather than per-tuple noise.

The interpretation requires the per-tuple support caveat. Because each ZEL028-2 tuple signature aggregates a median of 2 wells, individual tuple signatures carry substantially more sampling noise than the ZEL024 / HEK293 tuple signatures (median 13 wells). The headline numbers should therefore be read as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes the ZEL024 analysis surfaced (RNA processing and mitochondrial translation) rather than as a list of per-tuple definitive calls. Per-tuple confirmation requires deeper coverage on the specific tuple. The complementary single-position bb1 aggregation of the same ZEL028-2 wells (74 bb1 identities at minimum 500 wells in HEK293, 51 in A549 at minimum 500 wells, and 66 in H1650 at minimum 250 wells) is preserved as a higher-stability lower-resolution view in [paper5/tables/](#). ZEL031 is a two-position library, so its 3,261 (A549) and 3,556 (THP1) tuples at minimum 3 wells already represent its highest practical chemistry resolution. ZEL031 systems are reported at tuple resolution with the lower well-count threshold explicitly flagged.

Chemistry coverage across Z-Screen systems sets the analytical resolution per system: ZEL024 / HEK293 supports tuples at ≥ 10 cells, ZEL031 systems at ≥ 3 cells, and ZEL028-2 systems at ≥ 2 cells (v3 framing).



Supplementary Figure 2. Z-Screen chemistry coverage across systems. (A) Distinct tuples passing ≥ 2 , ≥ 3 , ≥ 10 , and ≥ 20 well thresholds in each system. ZEL024 / HEK293 supports tuple-level signatures at ≥ 10 wells, ZEL031 systems at ≥ 3 wells, and ZEL028-2 systems at ≥ 2 wells. (B) Distinct bb0+bb1 pairs passing 100 and 500 well thresholds in libraries that populate bb0; ZEL028-2 (which has no bb0) is now reported at full-tuple (bb1+bb2+bb3+bb4) resolution in panel A. ZEL024 supports both tuple and pair resolution; ZEL031 has too many distinct pairs for any single pair to carry meaningful well counts.

Cell-state transfer connects chemistry mapping across cell contexts (Figure 6)

The CRISPR atlases used here are dominated by K562 and THP1, while the densest combinatorial Z-Screen libraries in the canonical workspace are profiled in HEK293, A549, and H1650. Direct same-cell evaluation is therefore possible for THP1 (compounds and pair signatures) and partially for A549 (BAY-293 and other compounds). The remaining systems benefit from the cross-cell projection model developed in paper 4, which trains a centered ridge transfer between paired cell-line landmarks in the 32-dimensional scVI latent space.

Within the valid bridges in the present dataset, A549-to-THP1 transfer in ZEL031 reached transfer cosine 0.66 for bb1 programs and 0.59 for bb0; HEK293-to-H1650 and HEK293-to-A549 transfer

in the HEK293-rich libraries reached weighted cosines between 0.68 and 0.84, and ridge transfer improved on direct identity reuse in every paired bridge evaluated. The cross-cell transfer is reported in detail in paper 4; here it sets the boundary on what the CRISPR mapping can claim. The data support the claim that chemical programs are learnable and transferable across measured cell contexts, which extends the practical reach of the CRISPR mapping beyond the cell types where direct K562 or THP1 reference comparisons are possible.

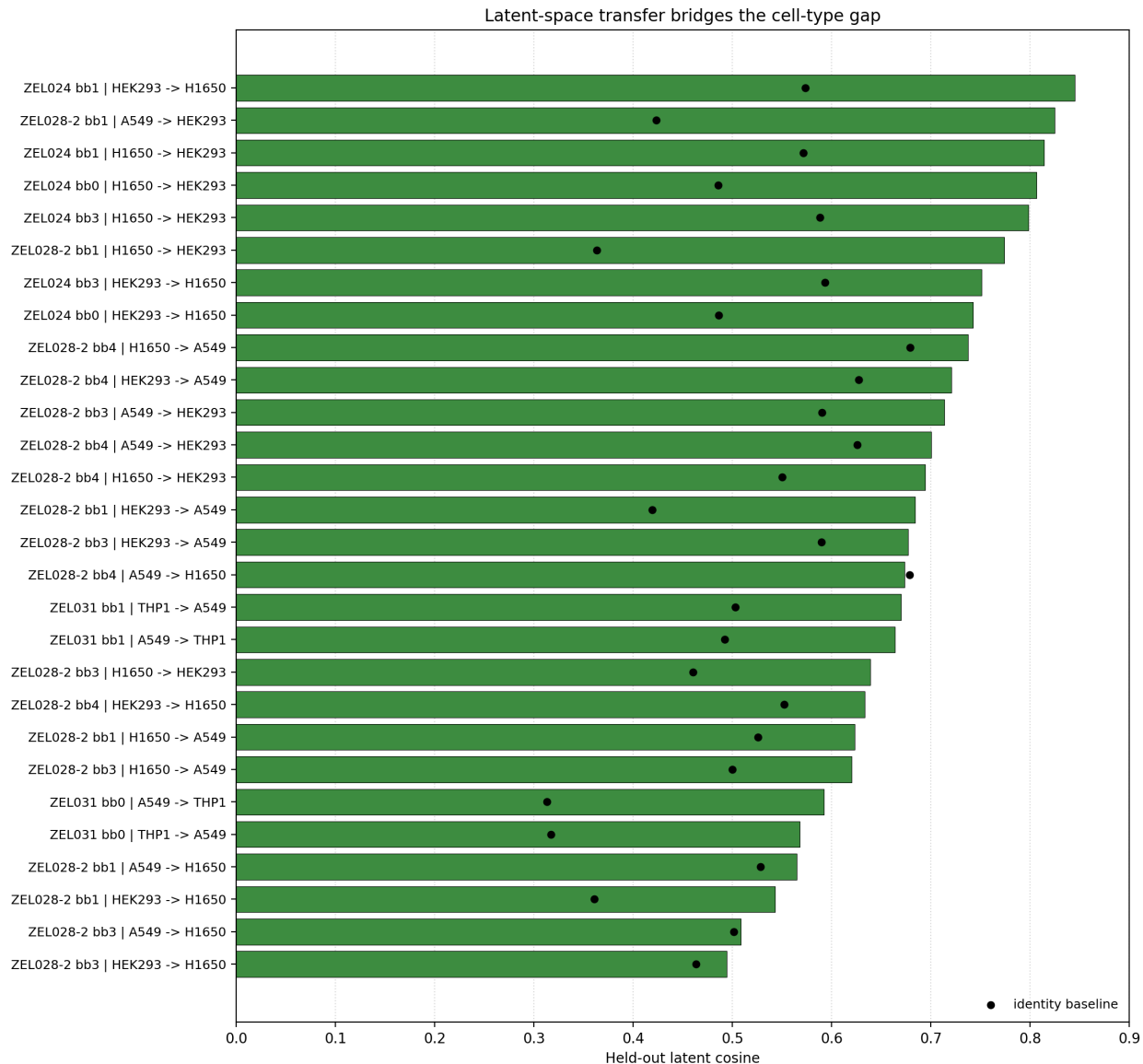


Figure 6. Cell-state transfer of Z-Screen chemical programs across cell contexts. The matched bridges support A549-to-THP1 transfer in ZEL031 and HEK293-to-A549 / H1650 transfer in the HEK293-rich libraries. Black points show direct identity baselines; green bars show learned ridge transfer performance.

Discussion

The result that frames the rest of the paper is the headline ZEL024 / HEK293 scan: 1,276 distinct chemical tuples each carry a calibrated CRISPR-like hypothesis at empirical $q = 0.05$, and the resulting hits cover 1,714 distinct CRISPR perturbation programs. The numbers themselves are not the central result; the central result is that each one of those tuples is a specific 4-position (bb0 / bb1 / bb2 / bb3) chemistry that a discovery team can synthesize, vary, and test, with the underlying CRISPR program available for orthogonal validation. The framework therefore turns Z-Screen output from a catalog of measurements into a structured search engine over genetic perturbation space.

The matched-cell controls validate the framework’s calibration. MZ1 recovers BRD4 with a clean separation against the same-cell THP1 CRISPR panel; STC-15 recovers IRF1 within the top two calibrated matches as a downstream proxy of METTL3 inhibition. Where the genetic analog is present, the framework recovers it; where the genetic analog is absent, the top hit is a documented downstream pathway rather than a generic stress signature. BAY-293 in A549 demonstrates the discovery layer: the framework surfaces a coherent V-ATPase / vesicular trafficking hypothesis ranked alongside, but not predicted by, primary pharmacology.

The combinatorial-logic test is the strongest evidence that the framework is reading real combinatorial chemistry rather than per-tuple noise. All 131 Norman doubles tested produce positive Spearman correlations between tuple-vs-double scores and tuple-vs-sum-of-singles scores. The chemistry follows the genetics’ combinatorial structure at chemistry-resolved level, which is the same structure the Norman atlas was built to express. That is not a result a one-profile-per-compound platform such as L1000 can produce, because there is no internal chemistry structure inside a single L1000 entry to test combinatorial logic against.

L1000 is also a useful concrete reference for data-quality positioning. Of the 47 named Z-Screen control compounds, 14 (29.8%) appear in the LINCS L1000 CMap LINCS 2020 build (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, AZD-3463, BMS-509744, BGT226, ZM-336372), and 3 of 4 Z-Screen cell lines (HEK293, A549, THP1) are present in L1000. That overlap was used to run an explicit cross-platform Spearman concordance test between Z-Screen up/down rank signatures and LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742), restricted to the 12,328-gene LINCS-visible universe so both platforms are scored in the same gene space. Across 11 (compound, cell_line) pairs with adequate coverage on both platforms, all 11 produced positive cross-platform correlation (median $\rho = 0.490$, mean = 0.393, 8 of 11 above $\rho = 0.2$). Five compounds reached empirical $p = 0.048$ against a same-cell-line unmatched-compound permutation null: crizotinib in A549 ($\rho = 0.679$), ZM-336372 in A549 ($\rho = 0.625$), rucaparib in A549 ($\rho = 0.530$), sorafenib in A549 ($\rho = 0.505$), and veliparib in A549 ($\rho = 0.490$). The HEK293 cell-line-pooled comparison, where sorafenib in HEK293T reached $\rho = 0.597$, separated matched from unmatched compound pairings at one-sided Mann-Whitney $p = 0.009$. Z-Screen rank signatures therefore recover compound-specific transcriptional structure that an independent reference platform also reports, in a gene-restricted concordance test that uses no platform-specific calibration. Full tables are in **Benchmarking/** in

this repository.

CRISPR similarity is not target identification. A small molecule may resemble a knockout because it directly engages the encoded protein, because it perturbs an upstream regulator, because it triggers a compensatory state, or because both perturbations converge on a broad stress program. The framework therefore treats every match as a state-level hypothesis until orthogonal validation upgrades it. The recurrent-neighbor diagnostic and the hub-strip analysis are reported alongside the calibrated headline numbers because broadly responsive CRISPR perturbations need to be flagged rather than silently lost in the leaderboard.

The framework is naturally tied to follow-up. The top combinatorial chemistries pointing toward a desired CRISPR state are concrete tuples with explicit per-position chemistry: bb0 / bb1 / bb2 / bb3 in ZEL024, bb1 / bb2 / bb3 / bb4 in ZEL028-2, bb0 / bb1 in ZEL031. A follow-up library can enrich the partner chemistry around those tuples, profile the new compounds, and update the model. The cross-cell transfer connection in paper 4 closes the remaining gap by mapping chemistry from a cell type that is dense in the screen into a cell type that carries the matched CRISPR reference.

A compact internal validation panel falls out of the analysis. BRD4 as a positive control; MED24, INTS1, EP300, FLI1, HDAC3, MYC, and SMARCD1 to test the recurrent transcription-and-chromatin programs; PDCD1LG2 for the strongest checkpoint-like hypothesis from the HEK293 BB-pair signatures; SMAD4 as a stress / state comparator; JAK2, IFNGR1, IFNGR2, and STAT1 to test the interferon-pathway proxies that emerged for STC-15 and several non-annotated discovery candidates. Running that panel in matched THP1 and HEK293 would directly test the two main claims: that known chemical-genetic analogs are recovered when present, and that the chemistry-resolved hypotheses replicate in a matched CRISPR context.

For drug discovery teams, the practical implication concerns the front end of a campaign. Instead of starting from a target and searching a separate compound database, a team can start from a desired CRISPR state and pull a ranked, calibrated list of combinatorial chemistries that move cells toward that state, with the chemistry decomposable into building-block coordinates that a medicinal chemistry team can act on.

Methods

Z-Screen data inputs

The analysis used the repaired canonical Z-Screen RNA-seq aggregate at `data/ZScreen_Canonical_Dataset/RNASeq`. Primary files: `Full_ANNDData_object_compressed.h5ad` (615,793 wells x 36,601 genes; rows are well-level pseudobulks of 5 to 10 cells per well, stored under the AnnData `obs` convention), `canonical_obs.parquet`, `canonical_chemistry.parquet`, and `canonical_scvi_latents.parquet`. The H5AD repair removed 140 wells affected by corrupted sparse-matrix chunks; the linked parquet files were aligned to the same `obs_index` after repair. See `DATA_REPAIR_REPORT.md`.

Rank signatures rather than expression vectors

The three CRISPR references and the Z-Screen RNA aggregate were generated on different platforms at different per-cell depths and with different normalization conventions. Forcing all of them into a common normalization risks introducing platform-specific artifacts that would then appear as similarity. Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of genes most up- and most down-regulated relative to background. The choice is conservative. It costs some statistical power compared with full expression-vector regressions, and provides the ability to compare across heterogeneous public datasets without assuming they were normalized the same way.

Z-Screen tuple signatures

For each library and cell line, building-block-annotation wells (`chemistry_grain == "building_block_annotation"`) were grouped by the concatenated tuple identifier `bb0_id|bb1_id|bb2_id|bb3_id` with unused positions filled as the literal string "NA" (so a ZEL024 tuple has the form `bb0|bb1|bb2|bb3|NA`, a ZEL028-2 tuple has `NA|bb1|bb2|bb3|bb4`, and a ZEL031 tuple has `bb0|bb1|NA|NA|NA`). Tuples with at least 10 wells were retained for the headline ZEL024 / HEK293 analysis (8,599 tuples, median 13 wells per tuple). A supplementary ≥ 20 -well threshold yielded 483 tuples and is reported alongside the headline as a stricter sensitivity check. ZEL031 systems used a ≥ 3 -well threshold (3,261 in A549, 3,556 in THP1) because the two-position library is too diverse for higher thresholds. ZEL028-2 systems used a ≥ 2 -well threshold because per-tuple coverage in that library does not support the ≥ 10 -well or ≥ 3 -well thresholds; under ≥ 2 wells, 4,041 (HEK293), 1,728 (A549), and 765 (H1650) tuples qualify, with a median of 2 wells per qualifying tuple. The shallower per-tuple support in ZEL028-2 is recorded in every signature's `n_cells` field (the field name follows the AnnData convention; the values count well-level pseudobulks), and is the reason the ZEL028-2 tuple analysis is interpreted as a chemistry-resolved leaderboard rather than as per-tuple definitive calls. For each tuple, target-well counts were aggregated by direct sparse-matrix slicing of the H5AD; the background was the same library and cell line excluding target-tuple wells. The effect vector was $\log\text{CPM}(\text{target}) - \log\text{CPM}(\text{background})$. Genes were ranked separately in the up and down directions; the top 250 per direction were retained.

The same-library, same-cell-line background was used in preference to a global baseline so that bias from chemistry-independent differences across libraries (e.g. the synthesis chemistry of one library affecting baseline cell state) does not appear as signal. Built by `paper5/scripts/run_tuple_signatures.py`.

Z-Screen BB-pair and single-position signatures

For systems where tuple-level coverage is thin or where a coarser-resolution anchor is useful, lower-resolution signatures with the same target-vs-same-library-background logic were also built. ZEL024 / HEK293 carries an auxiliary `bb0+bb1` pair anchor (84 pair signatures, each averaging ~ 165 distinct tuples) used in the resolution comparison only. ZEL028-2 (which has no substantive `bb0` position) carries a single-position `bb1` aggregation: under `paper5/scripts/run_bbpair_signatures.py`,

the `--pos-a bb0_id` flag falls through to `None` for ZEL028-2 wells, so the resulting signature aggregates all wells matching a given `bb1` identity regardless of `bb2`, `bb3`, `bb4`. The metadata for every signature records `pair_value_a`, `pair_value_b`, and the number of distinct tuples averaged in (`n_tuples_averaged`), so the level of chemical aggregation is explicit on every row. The ZEL028-2 single-`bb1` outputs are therefore retained as a higher-stability lower-resolution view of the same library; they aggregate hundreds of distinct downstream tuples per signature and are not chemistry-resolved at the per-tuple level. The main results for ZEL028-2 use the full-tuple analysis described above; the single-`bb1` outputs are preserved for inspection at `paper5/tables/ZEL028-2*_pair_bb0_id_bb1_id_min*` (the file naming reflects the original CLI invocation). Built by `paper5/scripts/run_bbpair_signatures.py`.

Public CRISPR rank signatures

Public CRISPR rank signatures used in this manuscript are bundled in `paper5/external_data/`:

- Replogle K562 genome-scale Perturb-seq, 8,603 perturbations [4].
- Norman K562 CRISPR-activation, 105 single + 131 paired perturbations [3].
- scPerturb harmonized THP1, 78 perturbations from Papalexi/Satija ECCITE-seq and Wessels/Satija combinatorial CRISPR-Cas13 [9].

The rank signatures were generated upstream from the source single-cell CRISPR data with `paper5/scripts/build_replogle_rank_signatures.py`, `paper5/scripts/build_norman_rank_signatures.py` and `paper5/scripts/build_scperturb_crispr_signatures.py`. Each builder converts the public single-cell perturbation data into the same up/down rank format used for Z-Screen, with control cells from the same dataset as background.

Rank-based chemistry-to-CRISPR comparison

Each rank signature was converted into a signed rank vector. Top up-genes received positive scores from $+(\text{top_k} + 1 - \text{rank}) / \text{top_k}$; top down-genes received the negative analog. Each Z-Screen signature was compared with each CRISPR reference signature by:

- mimic up-overlap (Z-up genes hitting C-up genes),
- mimic down-overlap,
- reverse up-down overlap,
- reverse down-up overlap,
- mimic total = mimic up + mimic down,
- reverse total = reverse up-down + reverse down-up,
- net mimic overlap = mimic total minus reverse total,
- rank cosine over shared top genes.

Comparisons used `top_k = 250` and recorded shared-top-gene counts per pair. To keep file sizes bounded for the 8,599-tuple by 8,603-CRISPR scan (74 million pair scores), only the top 50 CRISPR matches per Z-Screen query were retained as raw output. Rank cosine was reported as `NaN` when fewer than 10 genes were shared between the two signatures. Net mimic overlap (rather than rank

cosine alone) is the headline scoring metric because it penalizes anti-correlated matches, which would otherwise inflate cosine for compounds that produce broadly opposite-direction transcriptional states. Run by `paper5/scripts/run_chemistry_vs_crispr.py`.

Permutation-null calibration

For each top-K-per-query table the top 10,000 rows by net mimic overlap were calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature’s up-set and down-set from the dataset-specific gene universe. For each random draw the same mimic / reverse / net-mimic statistics were recomputed against the fixed Z-Screen up/down set. Empirical p-values were computed as $(1 + \text{count of nulls} \geq \text{observed}) / (1 + \text{n permutations})$ and converted to Benjamini-Hochberg q-values across the calibrated rows. The headline ZEL024 / HEK293 scan used seed 13 and 1,000 permutations per row.

A size-matched permutation null was used in preference to a parametric test because parametric distributions for rank-overlap statistics depend on assumptions (gene universe, top-k cutoff, signature size) that vary by reference. A permutation null preserves the size of each CRISPR signature exactly, so the same null procedure is valid across heterogeneous references without requiring a separate parametric calibration for each one. Run by `paper5/scripts/run_calibrate_matches.py`.

Recurrent-neighbor diagnostics and deduplicated headline tables

After calibration, per-CRISPR-signature recurrence tables were built that record how many distinct Z-Screen queries each CRISPR perturbation appeared as a top match for, both across the full top-K-per-query table and across the calibrated table at each q threshold. Two deduplicated views were also computed: one row per CRISPR program (best Z-Screen match by net z) and one row per Z-Screen query (best CRISPR match by net z). Finally, a hub-strip robustness summary records how many distinct programs and queries survive at $q = 0.05$ after dropping the top N most-recurrent CRISPR signatures (N in 0, 5, 10, 25, 50). The hub-strip analysis is reported alongside the headline rather than as a footnote because broadly responsive CRISPR perturbations are a known structural feature of large CRISPR atlases, and any chemistry-to-CRISPR framework that does not control for them will over-claim. Run by `paper5/scripts/run_recurrent_neighbor_diagnostics.py`.

Resolution comparison

For each ZEL024 / HEK293 tuple at ≥ 10 wells, single-bb3 signatures were built from the same library at ≥ 500 wells per bb3, with explicit n-tuples-averaged metadata. Single-bb3 signatures were compared with the Replogle K562 reference and the top 50 matches per query were recorded. For each tuple the top 5 Replogle CRISPR matches at tuple resolution, the matches the containing bb0+bb1 pair would produce, and the matches the containing bb3 single-position signature would produce were collected, and pairwise Jaccard overlaps were computed. The summary table reports the per-tuple overlaps and the aggregate distribution. Run by `paper5/scripts/run_resolution_comparison.py` plus the figure builder `paper5/scripts/make_resolution_figure.py`.

Norman combinatorial logic

For each Norman double-gene perturbation A+B (131 doubles), the constituent single-gene signatures A and B were located in the Norman metadata. Across the 8,599 ZEL024 / HEK293 tuples, the Spearman correlation was computed between the tuple-vs-double net mimic vector and the sum of the tuple-vs-single-A and tuple-vs-single-B net mimic vectors. The best Z-Screen tuple’s individual scores against the double and each single were also recorded, along with whether the best double score exceeded both single scores. Run by `paper5/scripts/run_norman_combinatorial_logic.py`.

Same-cell positive controls

The MZ1 and STC-15 positive controls used the matched-cell THP1 scPerturb reference. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its compound signature was built with `paper5/scripts/build_compound_rank_signatures_with_fallback.py` using a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). The same fallback was used for STC-15, STM2457, sorafenib, Crizotinib, and capmatinib in their respective sparse libraries. The fallback background is broader than a same-library vehicle control would be; the absolute z-score should be read in that context. The structure of the MZ1 result (BRD4 as the top match across permutation seeds) is robust. The internal-control summary integrating direct-target and proxy-target recoveries is in `paper5/tables/internal_control_summary_combined.csv`; the top-3 calibrated matches per same-cell THP1 compound are in `paper5/tables/second_positive_control_thp1_compounds_top3.csv`.

Cell-state transfer

The cross-cell projection used the 32-dimensional scVI latent coordinates from `canonical_scvi_latents.parquet`. For each library and cell-line pair with sufficient shared chemistry coverage, building-block-position profiles were averaged in latent space and a centered ridge transfer model was trained in cross-validation. Performance was reported as weighted row-wise cosine between predicted and held-out target profiles, with direct source reuse as the identity baseline. Run by `paper5/scripts/run_latent_transfer.py`. Detailed cross-cell results are reported in paper 4.

Limitations

The framework compares transcriptional states, not biochemical binding. A chemistry-to-CRISPR match is a state-level hypothesis until validated by orthogonal target engagement.

The MZ1 fallback background for matched-cell THP1 is broader than a same-library vehicle control would be. The structure of the MZ1 to BRD4 result is robust across permutation seeds; the absolute z-score should be read in the context of that broader calibration.

Local same-cell CRISPR coverage is THP1-dominant, with no HEK293 or HEK293T single-cell CRISPR atlas surfaced by the local public probe. The HEK293 chemistry results therefore use cross-cell K562 Replogle as the genome-wide reference; the cross-cell setting means the matches are chemistry-resolved CRISPR-like state hypotheses rather than direct cell-of-origin target calls.

Tuple-level chemistry resolution is dataset-dependent. ZEL024 in HEK293 supports 8,599 tuples at ≥ 10 wells with median 13 wells per tuple, the cleanest tuple-level system in the canonical workspace. ZEL028-2 systems use a relaxed ≥ 2 -well threshold because per-tuple coverage in that library does not support the ≥ 10 -well or even ≥ 3 -well threshold; the resulting tuple signatures aggregate a median of 2 wells, which makes individual tuple signatures substantially noisier than the ZEL024 / HEK293 tuple signatures. The ZEL028-2 tuple-level results are therefore interpreted as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes ZEL024 / HEK293 surfaced (RNA processing and mitochondrial translation) rather than as confirmed per-tuple calls; per-tuple confirmation would require deeper sequencing on individual tuples of interest. A complementary single-position bb1 aggregation of the same ZEL028-2 wells is preserved in `paper5/tables/` for users who prefer a higher-stability lower-resolution view. ZEL031 systems are tuple-resolution at lower well counts. The reporting tables annotate each signature with n-wells and n-tuples-averaged so the reader can directly judge the chemistry resolution behind any specific match.

Recurrent CRISPR perturbations exist; the recurrence diagnostic and hub-strip summary report how the headline counts change when the top hubs are dropped from the calibrated table. After dropping the 50 most-recurrent CRISPR signatures, 1,236 chemistry-resolved tuples and 1,664 distinct CRISPR programs still passed $q = 0.05$, so the diversity is not driven by hubs alone.

The targeted public scPerturb probe is laptop-scale (11 H5AD files under 500 MB each) and is not a comprehensive survey of all public HEK293 perturbation data.

Data availability

The full reproducibility package, including the canonical Z-Screen RNA-seq dataset, bundled public CRISPR rank signatures, all derived match tables, calibrated headline outputs, recurrence diagnostics, and figure-building scripts, is contained in this repository at `paper5/`. The repaired canonical RNA-seq aggregate is at `data/ZScreen_Canonical_Dataset/RNASeqAggregate/`. Public CRISPR rank signatures are bundled in `paper5/external_data/`. Their upstream source datasets are referenced below.

Code availability

All analysis and figure-generation scripts are in `paper5/scripts/`. The reproduction order, including dependencies on the bundled public CRISPR rank signatures, is in `paper5/README.md`. Package-level dependencies are listed in the repository root `requirements.txt`. The codebase was tested with Python 3.14.3 on Windows.

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